BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 304 (July 2024)

Digital Logic Design Laboratory

Final Project Report

Section: C1 Group: 01

8 Player Quiz Game Machine

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Academic Honesty Statement:

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1 Abstract

The game operates with a time limit for each question, managed by a timer circuit, ensuring players answer within a set period. Players' scores are tracked and displayed in real time, fostering competitive engagement. The system is designed to be simple, cost-effective, and scalable, allowing easy integration of more questions or features. The project offers a hands-on learning opportunity for those interested in basic electronics, circuit design, and interactive systems, making it an ideal tool for educational environments, social events, or as a hobbyist project.

This project highlights the fundamentals of game design, circuit building, and logic systems while providing an entertaining, competitive quiz experience.

2 Introduction

The 8 Player Quiz Game Machine is an interactive and engaging quiz system designed to facilitate group-based quizzes for up to eight players. The system consists of a central quiz unit, multiple input devices (buttons) for each player, and a visual output display for presenting questions, answer choices, and player scores. The machine operates without the use of a microcontroller, instead relying on standard electronic components such as switches and timers to control game logic and flow.

In this design, the central unit is responsible for managing the question bank and triggering questions and answers on the visual display. Each player interacts with the system by pressing a dedicated button to submit answers, and the system evaluates the responses through simple circuitry that compares the player's input to the correct answer. The system employs timer circuits to enforce time limits for responses, and visual feedback is provided via 7 segment display showing the current question, choices, and player standings.

The quiz machine supports multiple rounds, with each player's score being tracked and updated in real time. The use of standard electronic components, such as buttons for player input, switches for controlling game flow, and 7 segment displays for output, makes the design cost-effective and easy to build. This project is suitable for educational settings, social events, and entertainment purposes, providing a fun and competitive environment for group participation in quizzes and trivia games.

By eliminating the need for microcontroller programming, the system offers a more hands-on approach to understanding basic electronics and the principles of interactive game design.

3 Design

3.1 Problem Formulation

3.1.1 Identification of Scope

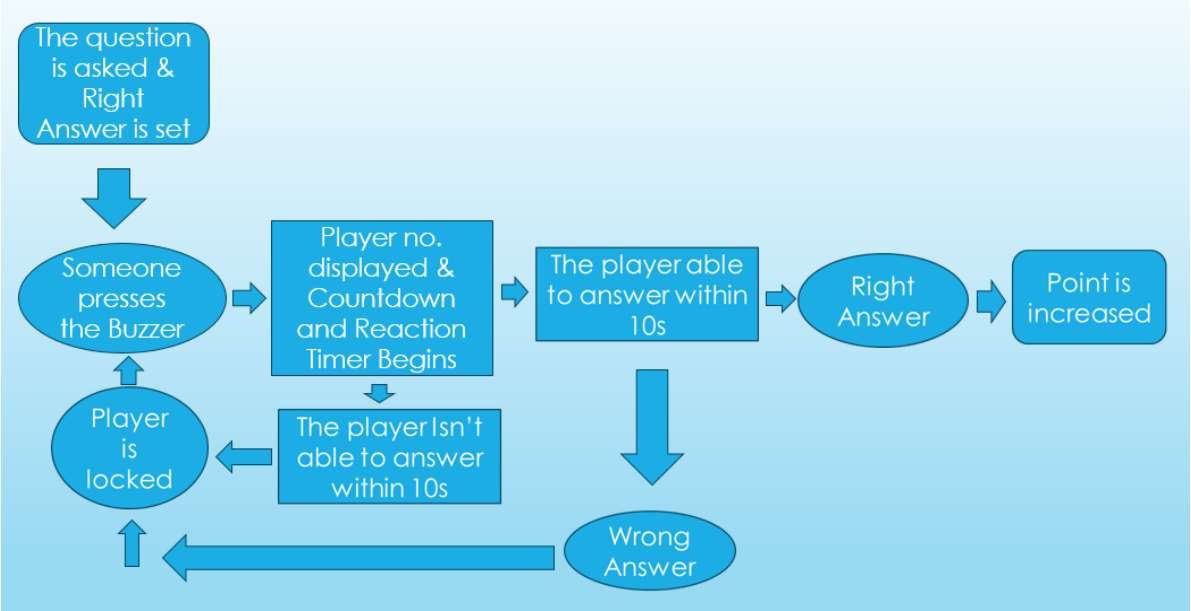
This project's scope is centered on the design, development, and deployment of an automated eight-player quiz game system that offers a fun and effective platform for quiz-based competitions in team-building, entertainment, and educational settings. Each participant can have their own buzzer, which automatically locks other buzzers so that only the first person to respond has a chance to answer. Setting the right responses and restarting the system for fresh questions are within the judges' authority. Correct answers cause a player's score to be automatically updated; false answers cause the player to be frozen, giving others a chance. The player number is dynamically shown on a seven-segment display, and prompt responses are guaranteed by a countdown timer. The system ensures dependable operation with features like debouncing and adaptability for future improvements by integrating several integrated circuits and logic gates. The goal of this project is to develop a reliable, scalable, and user-friendly system that satisfies societal demands while taking safety and environmental factors into account.

3.1.2 Literature Review

Innovations in digital electronics, interactive system design, and gaming technologies are all incorporated into the creation of an automated 8-player quiz game system. Due to their heavy reliance on human procedures for participant monitoring, answer timing, and scorekeeping, early quiz systems were prone to mistakes, delays, and inconsistencies. By enabling automated input handling, data storage, precise scoring, and timing functionalities, integrated circuits (ICs) such as the 74LS148 priority encoder, 74LS373 latch, 74LS138 decoder, and CD4510 up counter, which were made possible by the advancement of digital circuits, revolutionized quiz systems. In order to provide accurate and clean buzzer response detection, debouncing techniques—such as RC circuits in conjunction with NOT gates—became crucial for removing signal noise brought on by mechanical switches. Logic gates like the CD4075 OR gate and 74HC11 AND gate increased system reliability by controlling resets, locks, and conditional operations, while countdown timers, powered by parts like the 555 timer and timed ICs, improved fairness by enforcing stringent response windows.In order to enhance the user experience, contemporary quiz systems prioritize features like dynamic displays, automatic player locking, and fluid gaming. Because of its modular nature, they can be expanded in the future to incorporate more players, wireless capabilities, or improved scoring systems. Furthermore, environmental issues are addressed by energy-efficient designs and compact layouts, and operational robustness is ensured by the use of dependable ICs. This project combines these tried-and-true methods with creative ideas to produce an intuitive, scalable, and effective quiz game system that satisfies the requirements of modern apps while providing accurate and captivating gaming.

3.1.3 Formulation of Problem

This is our overall problem formulation in the form of flowchart.



- 8 player will participate.
- Judges will set the correct answer.
- Player's score will increase automatically for each right answer.
- Player will be locked out for each wrong answer and other players will get the chance for answering

3.1.4 Analysis

Here we can see, first of all question is asked by the judge and he will set the right answer also. 8 player will participate and they will press the buzzer. Player who pressed the buzzer first will be displayed in the 7 segment display. Countdown and reaction timer will start counting giving the player only 10 seconds to answer the question. If he fails to answer, he'll be locked out and new round will happen for that question without him. But if he could answer within 10 seconds and gives wrong answer, he will again be locked out and new round will begin again without him. Only when he gives the right answer within 10 seconds, his point will increase automatically.

3.2 Design Method (PO(a))

The main features of our project that we had considered while designing is-

❖ **Buzzer:** Each player will be provided a buzzer machine. When question is asked only the first player who pressed the buzzer will get the chance for answering the question. Other players buzzer will be locked out.

❖ **Display:** The player who pressed the buzzer first will be displayed. If he answers wrong, display will reset and show the new players number whoever presses the buzzer except him.

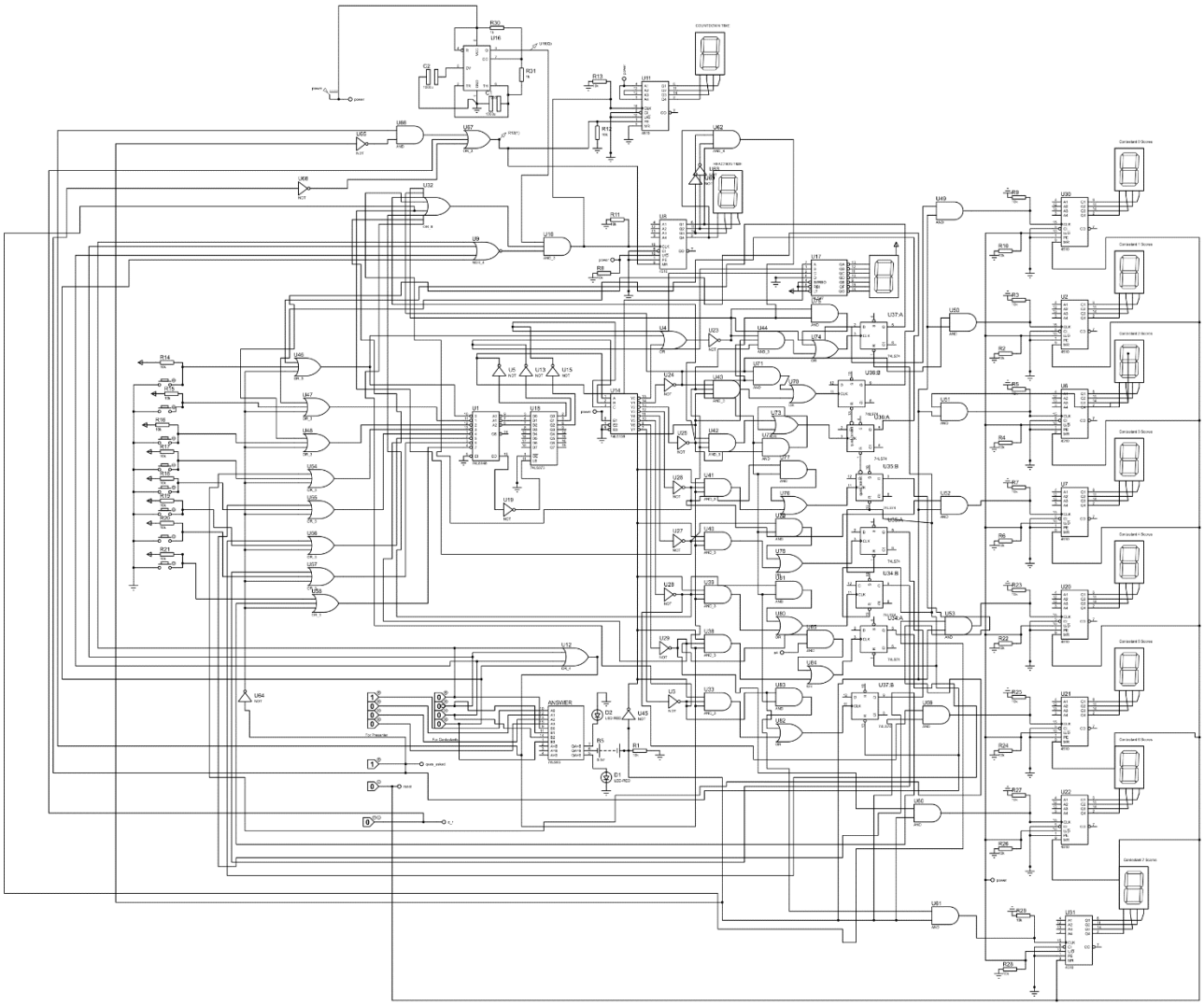
❖ **Countdown Timer:** When a player presses the buzzer, the countdown timer will start counting down from 9 seconds. The player who pressed the buzzer first will get 9 secs to answer the question. This timer will stop counting down when answer is correct. For each wrong answer the timer gets reset and start new round for that question.

❖ **Reaction Timer:** This timer will count the time required for the player to answer the question. If a player answer correctly for a question, it will stop counting. But if a player answers incorrectly, it will reset and new round for that question will begin.

❖ **Answer Evaluation:** There will be 4 options for each question. 4 buttons for answering will be provided to the players. 4 buttons will be provided to judge who will set the correct answer. When a player presses the right button, his score increases. When a player presses the wrong button, he gets locked out and new round for that question begins for other players.

❖ **Score Evaluation:** The player's score will increase automatically who gave the right answer. For wrong answer, he gets freeze and among other players who give right answer, his score will increment.

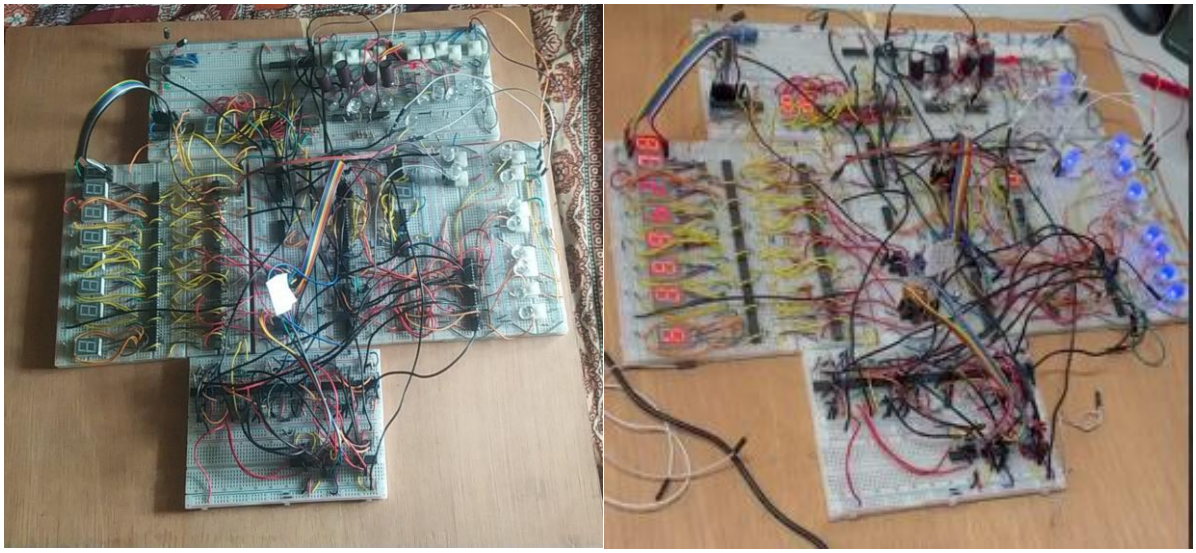
3.3 Simulation Model



Simulation Model Link-

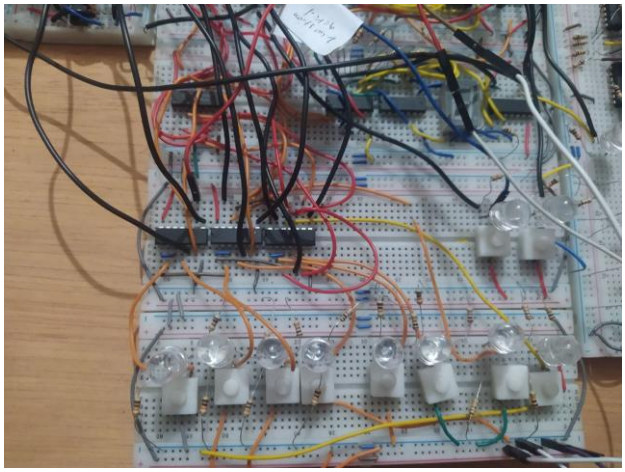
https://drive.google.com/file/d/10j6XEYWkPcQuAWhgnF4NZ_B1CdHLTa18/view?usp=sharing

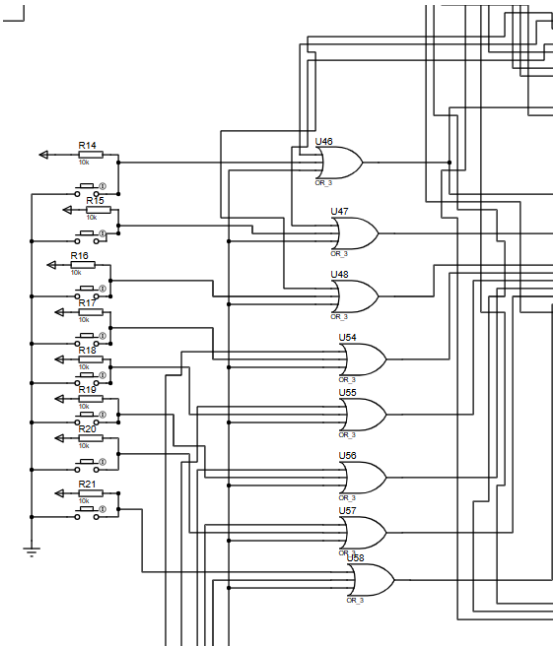
3.4 Hardware Design



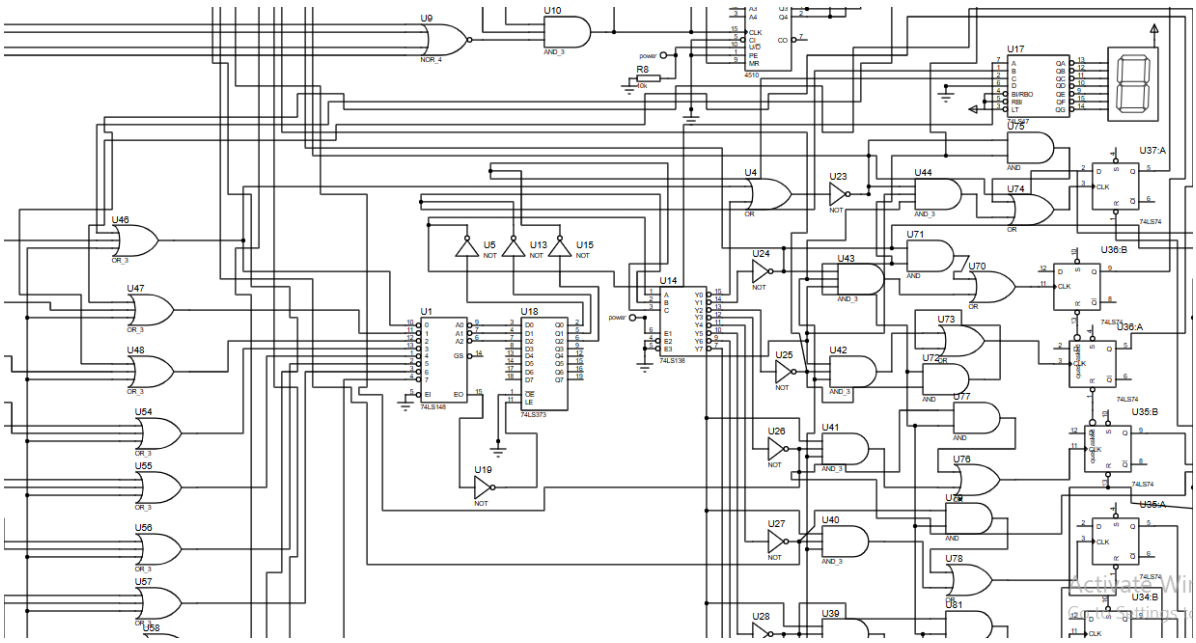
4 Implementation

4.1 Description



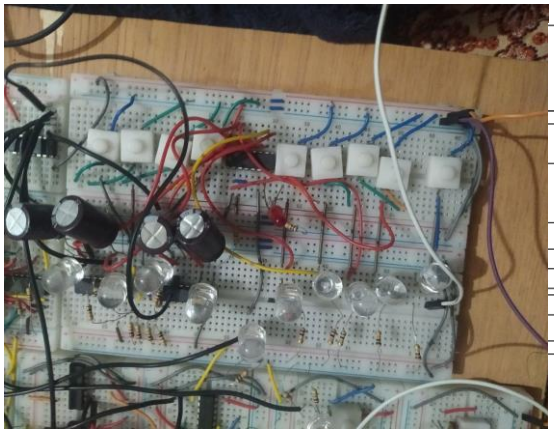


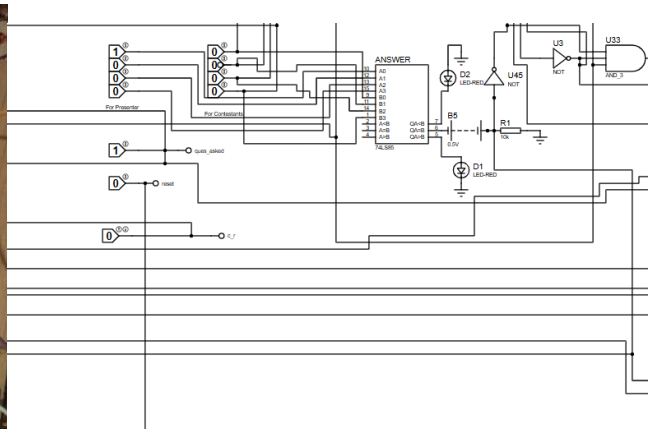
- **Buzzer:**
  1. CD4075 3 input OR Gate is used.
  2. One input Comes from D flip-flop which keeps the track of whether someone answered wrong in previous round or couldn't answer within 10 seconds.
  3. One input comes from buzzer button provided to players.
  4. Last input comes from whether question is asked or not.



❖ **Display:**

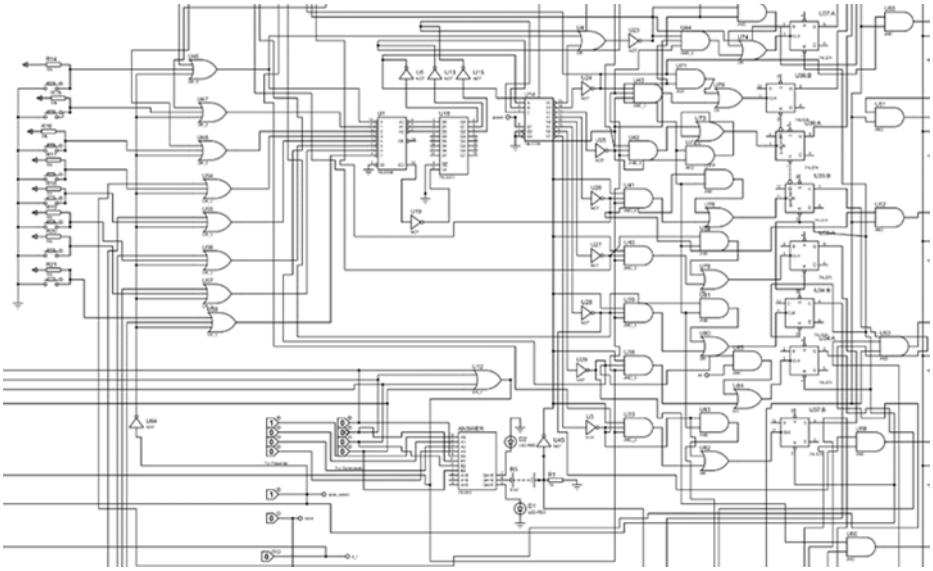
1. 74ls148 Priority Encoder decides who pressed buzzer by taking inputs from the outputs of 3 input OR gate.
2. 74ls373, 8 bit register stores this data
3. 74ls138 decoder decodes the stored data
4. 74ls47 bcd driver drives the 7 segment display to show players number.

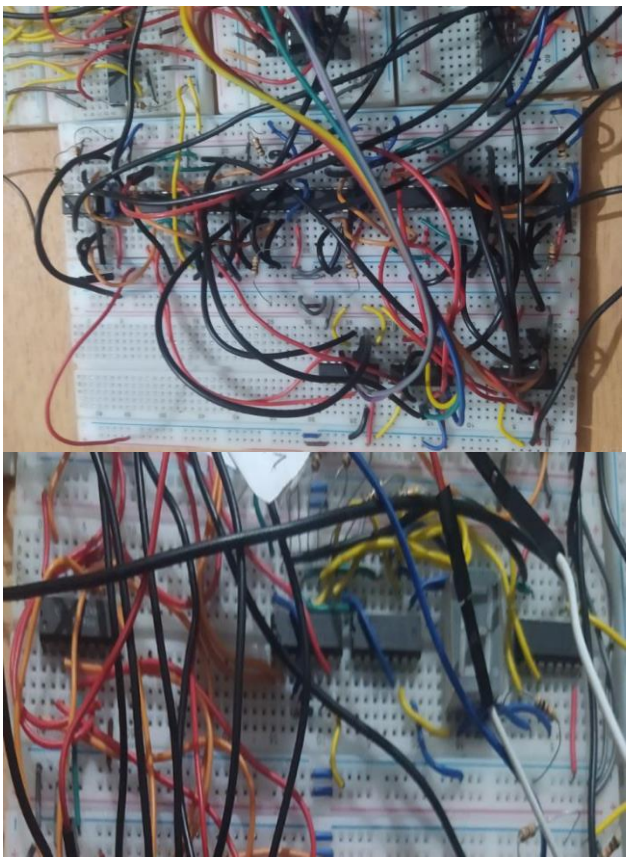




❖ **Answer Evaluation:**

1. There will be 4 options for each question.
2. 4 buttons for answering will be provided to the players and 4 buttons will be provided to judge who will set the correct answer.
3. Here 74ls85 comparator ic determines the correct answer.





❖ **Locking Player**

1.Clock will be given in d flip flop if player gives wrong answer or player doesn't give answer within 10 sec

2.Output of not gate will be 1 if any player presses buzzer,

3. Output of Or4 ic will be 1 if a player gives answer .If the answer is wrong output from the comparator after not gate will be 1.

3.Second input of and 2 gate comes from the output of timer which will be 1 if 10 sec is over.

4. So if any conditions of the two is true result of or gate will be 1.So the output from d flipflop will be 1 which will further go to the or3 gate and the player will be locked.

5 Design Analysis and Evaluation

5.1 Novelty

This 8-player quiz game system is distinctive because it seamlessly combines cutting-edge digital electronics with creative design elements to guarantee high accuracy, automation, and user engagement. This project uses ICs such as the 74LS148 priority encoder, CD4510 up clock, and 74HC11 AND gate, which offer real-time and error-free response processing, to implement automated scoring and locking procedures in contrast to conventional quiz systems that depend on manual processes. The automatic countdown and reaction timers to ensure fairness, the dynamic locking of buzzers to prevent simultaneous inputs, and the modular architecture that permits future improvements are some of the key innovative elements. Debouncing techniques guarantee precise and clean input signals, which is a significant advance over traditional systems that are susceptible to signal noise. A further layer of complexity is added to the action by incorporating a seven-segment display for player numbers and a mechanism that dynamically resets for wrong replies. The project also prioritizes scalability and energy efficiency by utilizing low-power components, which allows it to be modified for other quiz formats and player capacities. This project establishes a standard for the development and use of interactive quiz systems by solving common flaws in conventional systems and providing cutting-edge features.

5.2 Design Considerations

5.2.1 Considerations to public health and safety

By making sure that low-voltage and low-power components are used, the 8-Player Quiz Game System's design puts public health and safety first, lowering the possibility of electrical risks. In accordance with safety regulations, every circuit is encased in an insulated shell to avoid unintentional contact with active connections. By removing erroneous triggering, debouncing algorithms guarantee seamless gaming and prevent annoyance or confusion. Participants are guaranteed a stress-free and enjoyable experience because the clocks and locking mechanisms are made to work flawlessly, encouraging fairness and minimizing conflicts. The system reduces player and operator hazards by using dependable components and following sound design principles.

5.2.2 Considerations to environment

The design of the 8-Player Quiz Game System incorporates environmentally friendly practices by utilizing low-power digital ICs and components to minimize energy consumption during operation. The use of durable and long-lasting components reduces

electronic waste by ensuring the system's longevity and reliability. The materials selected for the enclosure are lightweight, non-toxic, and recyclable, further reducing the environmental impact. Additionally, the modular design allows for easy upgrades and repairs, extending the system’s lifespan and reducing the need for complete replacements. By focusing on energy efficiency and sustainability, the project contributes to reducing its overall environmental footprint.

5.2.3 Considerations to cultural and societal needs

By encouraging openness, participation, and equity in competitive settings, the project is carefully crafted to meet cultural and societal demands. Because it promotes collaborative learning and entertainment, it can be used in a variety of cultural contexts, such as corporate training sessions, community gatherings, and schools. The system promotes values that are universally valued, such as healthy competition, teamwork, and critical thinking. Accessibility for people of various ages, skills, and backgrounds is guaranteed by its versatility and user-friendly interface.
It is also a flexible tool that may be used in various locations because of its modular architecture, which enables customization to include culturally appropriate themes, questions, or languages. The approach fosters social contact and knowledge exchange, which improves community relationships and advances social and educational advancement. Its usability in a range of settings, from casual get-togethers to professional contests, is further enhanced by its portability and simplicity of setup. By taking these factors into account, the project satisfies the demand for inclusive and interesting technical solutions in society.

5.3 Investigations

5.3.1 Design of Experiment

The 8-Player Quiz Game System's experiment design was centered on evaluating each subsystem's performance, dependability, and functionality under many circumstances. To methodically assess specific parts and how they fit into the larger system, the trials were broken up into several phases.

To guarantee precise detection and display of the first player pushing the buzzer, the priority encoder (74LS148), latch (74LS373), decoder (74LS138), and display driver (74LS47) were initially tested. The efficacy of the debounce circuit in removing erroneous signals from the buzzers was assessed using an RC network. Timers were evaluated for countdown accuracy and response time when players responded to questions inside the given 10-second time limit. They were implemented using the CD4510 and controlled by the 555 timer and 74HC11 AND gate. In order to verify that players were properly locked out for incorrect answers and that scores increased for correct responses, the scoring system was verified by integrating the CD4510 up counter and comparator (74LS85). The CD4075 OR gate's reset capabilities were also assessed in a variety of reset scenarios, including timeouts, judge inputs, and incorrect replies.

To verify the system's robustness and dependability under extended operation, stress tests were performed by mimicking several quick buzzer presses and consecutive rounds. The system's capacity to manage real-time inputs, stop buzzers from activating simultaneously, and preserve fairness through automatic locking and scoring methods were all validated by the studies. Every testing phase confirmed that the system's architecture satisfies functional and performance specifications.

5.3.2 Results and Analysis

According to the testing results, the 8-Player Quiz Game System functions dependably and effectively in a variety of scenarios. The first player to press the buzzer was correctly detected by the priority encoder (74LS148), and the seven-segment display appropriately displayed the player number that corresponded to that identification. By successfully removing erroneous signals from mechanical switches, the debounce circuit ensured accurate buzzer detection free from errors or delays. With the help of the CD4510 and 555 timer, the timing mechanism precisely counted down from nine seconds and reset when certain conditions were met, such incorrect replies or a timeout. The 74LS85 comparator and CD4510 up counter are integrated into the scoring subsystem, which regularly updates scores for right responses and appropriately locks out players who provide incorrect answers.
The system's goals of automated scoring, real-time response handling, and games fairness were all met, according to the report. High operational efficiency was shown by the minimal latency seen in input detection and output display. Overall, the outcomes confirmed the system design's efficacy and its potential for real-world uses in recreational and educational contexts.

5.4 Limitations of Tools

Although the 8-Player Quiz Game System effectively achieves its goals, testing and development revealed some limitations with the tools and components used:

Limited Resolution of Timers: Although the 555 timer is inexpensive and multipurpose, it is not very accurate for time-sensitive tasks, particularly when triggered quickly.

Debouncing Accuracy: Although the RC circuit for debouncing works well for straightforward applications, it may not be able to withstand high speeds or significant mechanical noise.
Complexity of IC Interfacing: As the system grows, it may become more difficult to integrate several ICs, such as 74LS148, 74LS373, and CD4510, without careful design to maintain signal integrity and correct synchronization.

Scalability: Although the system is built for eight players, adding more would necessitate redesigning the architecture and adding more ICs, which would raise the complexity and expense.
Hardware in its physical form Limitations: Long-term durability and portability are hampered by problems including the deterioration of mechanical buttons and the circuit

board's fixed size.

These restrictions point to areas that could use improvement and possible enhancements to improve the system's scalability, performance, and efficiency.

5.5 Impact Assessment

5.5.1 Assessment of Societal and Cultural Issues

Through encouraging interactive learning and healthy competition in a range of contexts, including schools, community gatherings, and professional workshops, the 8-Player Quiz Game System has a beneficial social and cultural impact. It promotes involvement, critical thinking, and teamwork—skills that are universally admired.

Because of its flexibility, the system may be tailored to local languages, topics, or cultural preferences, which makes it appealing to a wide range of populations. It closes educational gaps and promotes involvement from people of all ages, skill levels, and cultural backgrounds by offering an equitable and inclusive forum for enjoyment and knowledge exchange. Additionally, the system fosters cooperation and cross-cultural interactions, strengthening social links and advancing inclusiveness and respect for one another. Additionally, its design minimizes prejudices and promotes confidence among judges, viewers, and participants by guaranteeing fairness and transparency.

5.5.2 Assessment of Health and Safety Issues

To provide a safe and comfortable user experience, the system was created with careful health and safety concerns. Participants of various ages, including young ones, can participate safely because the use of low-voltage components reduces the possibility of electrical risks. Additionally, the mechanical switches and buttons are made to withstand repeated use without resulting in harm or physical strain, guaranteeing ergonomic comfort even during prolonged gaming. By adhering to these safety measures, the system prioritizes the well-being of participants and spectators, making it a reliable and safe solution for use in educational, recreational, and community environment

5.5.3 Assessment of Legal Issues

In order to ensure that proprietary technologies are not violated, the components employed, including integrated circuits (ICs), timers, and registers, are commercially available and adhere to industry norms.To maintain ethical and legal integrity, the project does not use or duplicate patented designs without permission. The system is positioned as a valid and responsible invention that may be widely implemented by following certain legal frameworks.

5.6 Sustainability Evaluation

The 8-Player Quiz Game System was created with sustainability in mind, taking into account social, economic, and environmental factors.

Environmental Sustainability: To reduce energy consumption, the system makes use of energy-efficient parts including LED screens and low-power integrated circuits. A longer lifespan is ensured by using robust materials, which lowers the amount of electrical waste. Furthermore, the modular design makes upgrades and repairs simple, which encourages environmental responsibility even more.

Economic Sustainability: Because the project uses inexpensive parts and a straightforward circuit design, community centers and educational institutions may finance it. Its long-term economic viability is increased by its low operating costs and low maintenance needs.

Social Sustainability: The system promotes community involvement and cooperative learning by encouraging engaging and instructive gameplay. Because it encourages diversity, people of many ages and ethnic backgrounds can use it. Its acceptance in classrooms, seminars, and public gatherings is supported by its sturdy design, which guarantees dependable and safe use.

5.7 Ethical Issues

To guarantee equity, inclusion, and openness in its functioning, the 8-Player Quiz Game System was created with a significant focus on ethical issues. By automatically choosing the order of responses and excluding other players when a buzzer is pressed, the method ensures fair opportunity for all players and prevents bias.

By using dependable digital components, it is ensured that no player would unfairly benefit from technical mistakes. Additionally, the game steers clear of any potentially culturally inappropriate content or design, guaranteeing that it can be utilized in a variety of settings without violating social conventions or values. The project also conforms with intellectual property rules, guaranteeing that no patented technologies are used without authorization. The system preserves its integrity as an equitable, responsible, and inclusive educational instrument by adhering to these moral principles.

6 Reflection on Individual and Team work

6.1 Individual Contribution of Each Member

2006132- Software, Hardware Implementation

2006133- Software and Hardware Implementation

2006162- Software and Hardware Implementation

6.2 Log Book of Project Implementation

Week Passed	Milestone Achieved	Individual Role
5th week	**Software Implementation**	**133,162,132**
6th week	**Materials Collect**	**132,133,162**
8th week	**Hardware Implementation**	
10th week		
12th week	**Debugging**	
13th week	**Final Project**	

7 Communication to External Stakeholders

7.1 Executive Summary

An interactive tool that uses a competitive quiz format to improve learning is the 8-Player Quiz Game System. It uses digital elements including encoders, registers, and timers to ensure fluid gameplay and can support up to eight players, each of whom has a buzzer. In addition to tracking scores, the system shows the player number that buzzed first and bans out players who provide an inaccurate response. Answer time is limited to nine seconds by a countdown timer. For precise button pressing, the design incorporates an RC debounce circuit. The system, which prioritizes equity, safety, energy efficiency, and sustainability for long-term usage, is perfect for educational and community contexts.

7.2 User Manual

Welcome to the 8-Player Quiz Game System! This interactive quiz game is designed to enhance learning and foster healthy competition. The system supports up to eight players and includes features such as buzzer input handling, automatic score updates, and time management for each question.

Overview of the System

participants: There is room for eight participants, each of whom gets a buzzer and an opportunity to respond to questions.
Display: A seven-segment display shows the player number that pressed the buzzer first.
Timer: When a player pushes the buzzer, a 9-second countdown timer begins. If the answer is wrong or the allotted time has passed, the timer resets.Correct responses are automatically scored; incorrect replies cause the player to freeze, giving the other players an opportunity to answer.
Components: To control buzzer inputs, answer validation, and timing, the system makes use of integrated circuits (ICs) such as the 74LS148 encoder, 74LS373 register, and 74LS85 comparator in addition to other digital components**.**

Installation Instructions

1.Connect the system to a regulated DC power supply. The system requires a +5V DC input for proper operation. Ensure that the power supply provides adequate voltage and current as specified by the system's requirements.

2.Each player will have a buzzer button. The buzzers are internally connected to the system, so there's no need for external wiring

3.The player's number who presses the buzzer first will be displayed on the 7-segment display. This is automatically managed by the system and does not require any external setup.

7.3 YouTube Link

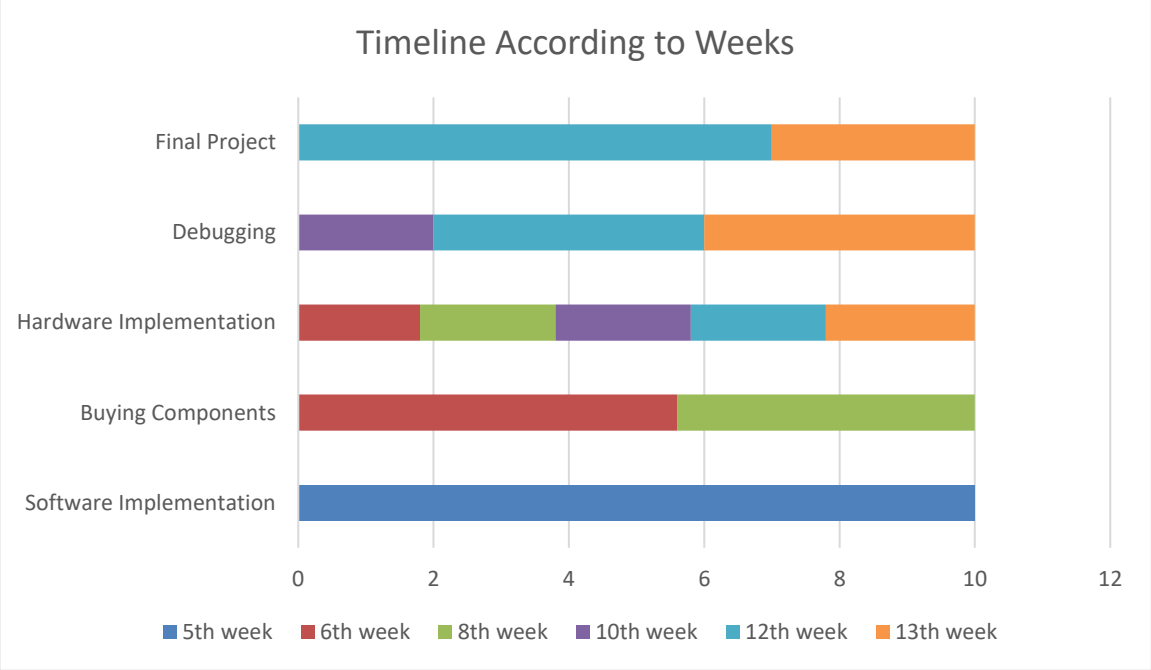
https://youtu.be/Xpo7z3hpue0

8 Project Management and Cost Analysis

8.1 Bill of Materials

Components	Price(tk)
Resistors (10k,1k)	200
Capacitors (2200uF,0.1uF,470uF)	550
Switch	150
ICs	950
7 Segment Display +Led	550
Jumper Wires	850
Cardboard	200
	Total-3450

8.2 Timeline of Project Implementation



9 Future Work

The 8 Player Quiz Game Machine, built without a microcontroller, provides a strong foundation for further development and enhancement. While the current design focuses on basic electronics, the following future improvements and expansions could significantly increase its functionality, scalability, and versatility:

1. **Incorporation of Advanced Displays:**
   - **Touchscreen Interface:** A touchscreen interface could replace buttons, making the system more user-friendly and allowing for more complex inputs, such as selecting from a list of categories or customizing the quiz.
2. **Wireless Connectivity:**
   - **Bluetooth/Wi-Fi Integration:** Adding wireless functionality would enable participants to connect their smartphones or tablets as input devices, eliminating the need for physical buttons. This would allow the system to be more flexible and scalable, accommodating larger groups or remote players.
   - **Remote Control:** Facilitating wireless control for the game host would allow better management of the quiz, such as selecting questions or controlling the game flow from a distance.
3. **Sound and Multimedia Features:**
   - **Audio Feedback:** Incorporating sound effects or voice prompts could make the game more engaging. For instance, correct or incorrect answers could trigger specific sounds, and additional audio cues could guide the players throughout the game.
   - **Video Integration:** The addition of video clips or animations related to the questions could enhance the visual appeal and interactivity of the quiz, making it more immersive for participants.
4. **Expandable Question Bank:**

- **Modular Question Storage:** The system could be upgraded to support a larger, modular question bank, either through SD cards or external memory modules. This would allow for the addition of a wide range of topics, levels of difficulty, and different quiz formats.
- **Category-Based Quizzes:** Future versions could include a selection menu to allow players to choose specific categories (e.g., history, science, sports), adding variety to the game.

5. **Player Customization and Profiles:**
   - **Player Profiles:** Adding the ability to create player profiles could allow tracking of scores and achievements over time, making the system more personalized and competitive.
   - **Multiplayer Support:** Future iterations could allow more than eight players to participate, either by adding more input devices or using wireless connections.
6. **Educational and Interactive Features:**
   - **Learning Mode:** A mode could be developed where questions provide explanations after each answer, turning the quiz into an educational tool that helps users learn from their mistakes.
   - **Timed Challenges and Multiplayer Tournaments:** Competitive game modes, such as a timed challenge where players race to answer, could be implemented, encouraging faster decision-making and critical thinking.
7. **Integration with Other Systems:**
   - **Online Leaderboards:** A cloud-based leaderboard system could be implemented, allowing players to compare their scores against others globally. This could be particularly useful for educational environments or public quiz competitions.
   - **Integration with External Quiz Platforms:** The system could be expanded to interface with existing quiz platforms, allowing quizzes to be conducted remotely and live with other online participants.

10 References

- **https://www.engineersgarage.com/8-player-quiz-buzzer/**
- **https://www.youtube.com/watch?v=mljXSRayRuA**